PT5053 Qualitative Research Designs in Healthcare



LUCY METCALFE

SENIOR LECTURER

L.METCALFE1@LONDONMET.AC.UK

Learning Outcomes

1. **Define qualitative research** and explain its purpose within healthcare and physiotherapy practice.
2. **Differentiate qualitative and quantitative research**, including underlying paradigms and types of questions each approach addresses.
3. **Identify common qualitative research designs** used in healthcare (e.g., phenomenology, grounded theory, ethnography, case study).
4. **Explain why qualitative approaches are valuable** for exploring experiences, meaning, and patient perspectives.
5. **Use the CASP tool** to recognise what makes a qualitative study methodologically sound.
6. **Interpret how qualitative research informs physiotherapy practice**, including patient-centred care and clinical decision-making.
7. **Understand expectations for Assessment 2**, including appraisal and the application of qualitative principles.


Subjectivity
Naturalistic setting
Qualitative Research
Contextualization
Inductive reasoning
Flexibility
Richness and depth
Interpretation and meaning -making

Qualitative Research Methods

- Uses inductive approaches to generate novel insights into phenomena that are difficult to measure quantitatively
- Qualitative methods can generate a comprehensive description of processes, mechanisms or settings
- They can characterise participants PERSPECTIVE and EXPERIENCES in great depth
- Qualitative researchers are concerned with the meaning people attach to thing in their lives.

Qualitative Research Methods

- Definition:

"Qualitative research consists of any kind of research that produces findings not arrived at by means of statistical procedures or other means of quantification. Instead, it focuses on the in-depth understanding of human experiences and social contexts." (Patton, 2015)

A strategy for systematic collecting, organisation and interpretation of textual information

Thoughtful --> Deliberate --> Intentional --> Systematic

Qualitative Research Methods

- Qualitative research is a type of research that explores and provides deeper insights into real-world problems.
- Instead of collecting numerical data points or intervening or introducing treatments, just like in quantitative research, qualitative research helps generate hypotheses to further investigate and understand quantitative data.
- Qualitative research gathers participants' experiences, perceptions, and behaviour. It answers the how's and whys instead of how many or how much.

Qualitative Research Methods

- Qualitative research involves collecting and analysing non numerical data (e.g. text, video or audio) to understand concepts, opinions or experiences
- It can be used to gather in depth insights into a problem or generate current ideas for research
- Qualitative research is commonly used in the humanities and social sciences

Qualitative Research Methods



UNDERSTAND HOW AND WHY



BEHAVIOURS,
EMOTIONS, EXPERIENCES, AND
BELIEFS

Key Differences Between Qualitative and Quantitative Researc

Aspect	Qualitative Research	Quantitative Research
Nature of Data	Non-numerical (texts, images, videos, observations)	Numerical (statistics, percentages, figures)
Objective	Explores in-depth meanings and patterns	Tests hypotheses and measures relationships
Approach	Subjective, interpretative	Objective, empirical
Data Collection Methods	Interviews, observations, focus groups, case studies, document analysis	Surveys, experiments, structured observations, secondary data
Analysis Methods	Thematic analysis, content analysis, discourse analysis	Statistical tests (t-tests, chi-square, regression analysis)
Research Design	Flexible and evolving during the study	Structured and follows a fixed plan
Sample Size	Small, purposive, non-random	Large, random, representative
Generalizability	Limited to specific contexts	High generalizability to larger populations
Researcher's Role	Active participant in interpretation	Detached, independent observer
Example Research Questions	"How do students experience online learning?"	"What percentage of students prefer online learning over in-person classes?"

Qualitative Research – Key Methods



Focus on understanding human behaviour and experiences



Use open-ended, exploratory questions



Collects data in natural settings (e.g. interviews or observations)



Emphasizes meaning, context and depth over numerical data



Involves subjective interpretation by researches



Thoughts

'Not everything that can be counted counts, and not everything that counts can be counted'

- Albert Einstein

What are we producing from qualitative research?

- Recurrent themes *
- Hypotheses *
- Survey instrument measures
- Conceptual models or theories *

What are themes?

- Identify and characterise:
  - Patterns of behaviour
  - Group interactions
  - Individual perceptions

Research Questions

1. Define the purpose of the research (knowledge gap)
2. Determine appropriate methods
3. Guide studying planning
4. Frame analysis and findings

Research Questions

- Inductive, exploratory
- Frame as a research question, aim, objective (no prior hypotheses)
- Focus on a single phenomenon, concept or idea

Writing a research question

- State the goal using a verb
- Identify topic of interest
- Use neutral or non directional language
- Define the sample and setting
- Be precise

Research Question

To understand interpersonal and systems-level factors relevant to delivering health care to emergency department patients who are homeless

How do patients with chronic musculoskeletal conditions experience the shift from in-person physiotherapy to digital or hybrid physiotherapy models?

How do physiotherapists make clinical decisions when working with patients who present with multimorbidity and fluctuating symptoms?

What factors influence patient motivation and engagement in self-management programmes within physiotherapy practice?

Research Question – PEO Framework

- Qualitative research, at its core, asks open-ended questions whose answers are not easily put into numbers, such as "how" and "why.
- Due to the open-ended nature of the research questions, qualitative research design is often not linear like quantitative design.
- The PEO framework is useful for building research questions that have a qualitative focus.

PEO Framework

P	Population and their problems • how would you describe the patient or population of interest?
E	Exposure • what has the population been exposed to? • What have they experienced?
O	Outcomes or themes • Outcome(s) of interest • unlike in PICO, the outcome is almost always used in the search terms with the PEO framework

PEO Framework

- Population and their problems: Nurses
- Exposure: Revalidation
- Outcomes or themes: Attitudes, feelings and views

- Example question: What are nurses attitudes towards revalidation?

Research Question: SPICE

- S – Setting (where)
- P - Perspective (for whom?)
- I – Intervention (what?)
- C – Comparison (compared with what?)
- E – Evaluation (with what result?

S	Setting - where?
P	Perspective - for whom?
I	Intervention - what?
C	Comparison - compared with wh
E	Evaluation - with what result?

SPICE Framework

SPIDER – Qualitative and mixed method studies

- S – Sample
- P – Phenomenon of interest
- D – Design
- E – Evaluation
- R – Research Type

SPIDER Question: mixed methods

S	Sample	Teen parents
PI	Phenomenon of interest	Pregnancy
D	Design	Questionnaire
E	Evaluation	Experiences
R	Research type	Qualitative

When to use Qualitative Research

- Exploring new topics
- Understanding cultural and social contexts
- Examining personal experiences and narratives
- Analysing group dynamics and interactions
- Studying change over time
- Investigating language and media
- When quantitative methods are insufficient
- Conducting participatory or action research

Qualitative Methodologies

- Phenomenology
- Ethnography
- Grounded Theory
- Case Study
- Narrative Research
- Action Research



Phenomenology

- Definition:
- Study of the meaning of phenomena or the study of the particular
- Investigate experiences from the individual's perspective
- Phenomenology is looking into the LIVED EXPERIENCE of the participants and aims to examine how and why participants behave in a certain way from their perspective

Understanding lived experiences and how individuals perceive a phenomenon

- Key Features:
- In-depth interviews to explore subjective experiences
- Bracketing to reduce researcher bias
- Open ended questionnaires

Ground Theory

- Definition:
- Generation of a theoretical model through the experience of observing a study population and developing a comparative analysis of their speech and behaviour
- Developing theories from systematically collected data
- Goal: explain how and why an event occurs or how and why people might behave a certain way
- Key Features:

Inductive approach – theories emerge from data (bottom up)

Constant comparison – data is analysed throughout.

Collecting data from interviews or observations

Analysing to identify patterns and themes

Examples

- How to patients develop chronic pain?
- Explore coping mechanisms?
- Cognitive Behaviour Therapy
- Medicine
- Requires multiple rounds of interviews
- Aims: theory generation in under researched areas
- Can be time intensive due to multiple rounds

Ethnography (understanding through experiences)

- Ethnography as a research design originates in social and cultural anthropology and involves the researcher being **immersed** in the participants environment.
- Through **immersion**, the ethnographer can use a variety of data collection techniques to produce an account of the social phenomena
- Researches aim **immerse** themselves into the research population and come out of it with accounts of actions, behaviours, events etc
- Direct involvement of the research with the target population is one benefit of ethnographic research
- Involves observing participants in their natural environment
- Can use in depth interviews to understanding

Case Study

- Definition:

In-depth analysis of a single case or small group within a real-life context.

Investigate a single individual or group of individuals to gain an in depth understanding of their experiences, behaviours, or outcomes

- Key Features:

Uses multiple data sources (e.g., interviews, documents).

Can be exploratory, explanatory, or descriptive.

Narrative Research

- Definition:

Collecting and analysing life stories and personal experiences.

Reporting on qualitative research involves including details and descriptions of the setting involved and quotes from participants

- Key Features:

Chronological or thematic storytelling analysis

Action Research

- Definition:

A collaborative approach where researchers and participants work together to solve a problem.

- Key Features:
- Cyclical process: Planning → Action → Observation → Reflection.
- Stakeholder involvement for real-world impact.

Data Collection Methods

- Observations
- Document and content analysis
- In depth interviews
- Focus group discussions
- Diaries and Journals
- Case Studies

In Depth Interviews

- Guided conversation
- Qualitative interviews are one of the most widely used methods in qualitative research
- They involve direct interaction between a researcher and a participants to gather detailed information about individual experiences, beliefs and perspectives
- Interviews can take place in person, phone, video
- Depending on the studys aims, they may follow a structured format with pre determined questions, a semi structured format with flexible prompts or unstructured that allows open conversation
- Interviews are effective for exploring personal narratives and understanding emotional topics

Sampling for in depth interviews

- Aim is to identify 'key informants'
- Goal is breadth not representativeness
- Size varies depending on the complexity of the inquiry
- Determined by theoretical saturation (point at which no new concepts emerge from the data)

Interview Guides: Data Collection

- Framework for the interviewer
- List of main questions and probes
- Open and non directive (non directional / explorative)
- Interviewer may diverge or move away from questions to explore themes and areas
- Interviewers may reword questions or reframe questions

In Depth Interviews - Interview

- Good rapport is imperative
- Participants will talk candidly only if they
- Feel comfortable
- Trust the interviewer
- Feel secure about confidentiality
- Believe the interviewer is interested in their story
- Does not feel judged

Focus Groups

- Focus groups are most useful when the goal is to understand how opinions form and evolve through discussion.
- They are effective for exploring collective experiences, testing concepts or materials, and identifying themes that reflect group consensus or disagreement.
- Group of people with certain characteristics generate narrative data in a focused discussion

Observations

- Observations involve systematically watching and recording behaviours, interactions, or events as they occur in their natural context.
- This method helps researchers understand how people act and interact in real situations rather than relying only on self-reported accounts.
- Observations can be structured, where the researcher follows a specific checklist or framework, or unstructured, allowing events to unfold naturally.
- They are valuable for studying patterns of behaviour, social dynamics, and environmental influences that participants themselves may not articulate in interviews or surveys.

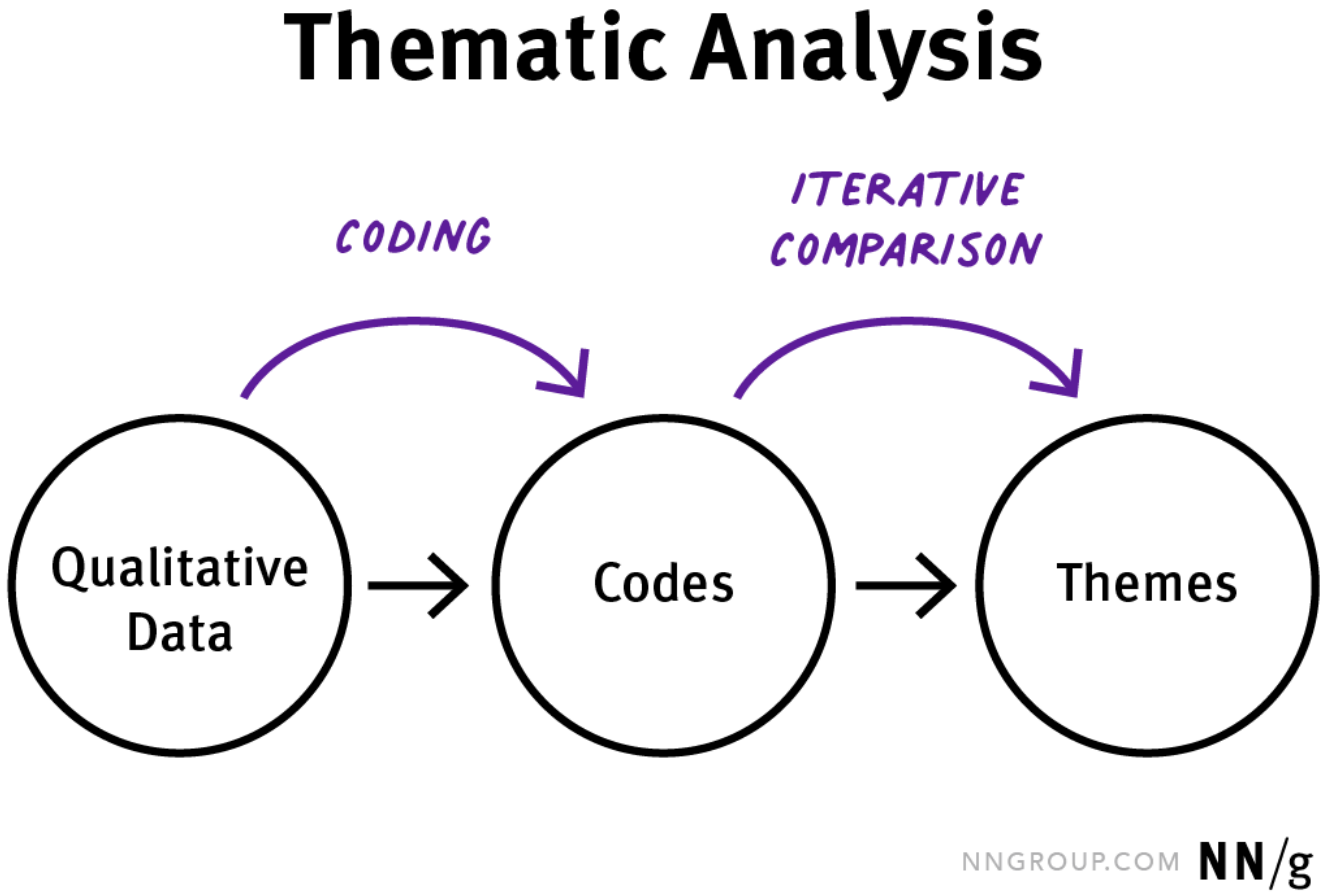
Case Studies

- Case studies involve an in-depth examination of a single case or a small number of cases within their real-life context.
- A “case” can refer to an individual, group, organization, community, or event.
- This method provides a holistic understanding of complex issues by combining multiple sources of qualitative data, such as interviews, observations, and document analysis.
- Case studies are particularly useful when researchers aim to describe, explain, or interpret phenomena that cannot be separated from their surrounding circumstances.

Document and Content Analysis

- Document and content analysis is best used when researchers aim to understand how meaning is constructed in texts or other communicative materials.
- It is appropriate for studies focused on discourse, representation, or cultural values as reflected in language and imagery.
- This method is also suitable when time, access, or ethical considerations make direct data collection from participants impractical.
- Researchers often use it to complement interviews or observations, providing background context or evidence to support other findings.

Data Analysis



Data Analysis



Thematic Analysis



Content Analysis



Narrative Analysis



Discourse Analysis

Key Characteristics



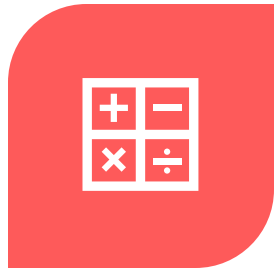
NATURALISTIC INQUIRY



INDUCTIVE APPROACH



SUBJECTIVE AND INTERPRETATIVE ANALYSIS



USE OF NON-NUMERICAL DATA

What are codes?

- Word of short phrase that represents the essence or key attribute of narrative / verbal information
- Used to codify or categorise the data
- Coding is the process of organising the data into chunks that are alike
- Codes are developed into a code structure

What is code structure?

- Compilation of the emerging codes
- Can evolve through the analysis

Pros and Cons:

Pros:	Provides rich, detailed insights.	Captures social and cultural contexts.	Allows flexibility in exploring emerging ideas.
Cons:	Time-consuming and labour-intensive.	Difficult to generalize findings and reproducibility.	Requires careful interpretation to minimise researcher bias

Assessment 2 – Qualitative



Assignment 2: Critical Appraisal of Qualitative Research



RESEARCH ARTICLE Open Access

Exploring expectations and perceptions of different manual therapy techniques in chronic low back pain: a qualitative study

Check for updates

A. Plank[1], A. Rushton[1,2], Y. Ping[1], R. Mei[1], D. Falla[1] and N. R. Heneghan[1*]



Open access Original research

BMJ Open

Lived experiences of recovery following musculoskeletal trauma 6 months following injury in the UK: a qualitative study

Nicola Middlebrook ,[1] Nicola R Heneghan ,[2] Maria Moffatt ,[3] Lucy Silvester ,[4] Deborah Falla ,[5] Alison B Rushton ,[6] Andrew Soundy [2]

LONDON METROPOLITAN UNIVERSITY

School of Human Sciences – Health Sciences
Template for PT5053 Assignment:

Introduction: **Approx 100 words**	
Study Design **Approx 120 words**	
Sampling and Recruitment: **Approx 120 words**	
Data Collection Methods: **Approx 120 words**	
Data Analysis and Findings **Approx 150 words**	
Strengths, Limitations and Trustworthiness **Approx 100 words**	
Relevance to Physiotherapy Practice **Approx 100 words**	
Conclusion **Approx 40 words**	

Student Template

- Use the student template to complete assignment
- Attach qualitative CASP as appendices

CASP – Qualitative

Section A Are the results valid?	
1. Was there a clear statement of the aims of the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *what was the goal of the research?* • *why was it thought important?* • *its relevance*	
2. Is a qualitative methodology appropriate?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If the research seeks to interpret or illuminate the actions and/or subjective experiences of research participants* • *Is qualitative research the right methodology for addressing the research goal?*	
3. Was the research design appropriate to address the aims of the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *if the researcher has justified the research design (e.g., have they discussed how they decided which method to use)*	
4. Was the recruitment strategy appropriate to the aims of the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If the researcher has explained how the participants were selected* • *If they explained why the participants they selected were the most appropriate to provide access to the type of knowledge sought by the study* • *If there are any discussions around recruitment (e.g. why some people chose not to take part)*	

CASP Section A – Research Aims

Was there a clear statement of the research aims?

- Did the study clearly define what it was investigating and why?

Is a qualitative methodology appropriate?

- Does the study use qualitative methods (e.g., interviews, focus groups) to explore lived experiences and perceptions?

Was the research design appropriate?

- Was the chosen method (e.g., ethnography, case study, grounded theory) suitable for answering the research question?

Section A: Research Aims

- Was the recruitment strategy appropriate?
- Were participants relevant to the study?
- Were inclusion/exclusion criteria well defined?
- Was the data collected in a way that addressed the research issue?
- Were interviews, focus groups, or observations conducted rigorously?
- Were data collection methods clearly explained?
- Has the relationship between the researcher and participants been adequately considered?
- Were researcher biases and influences acknowledged?
- Did the researchers reflect on how their role affected the data collection?

CASP Section B: What are the results?

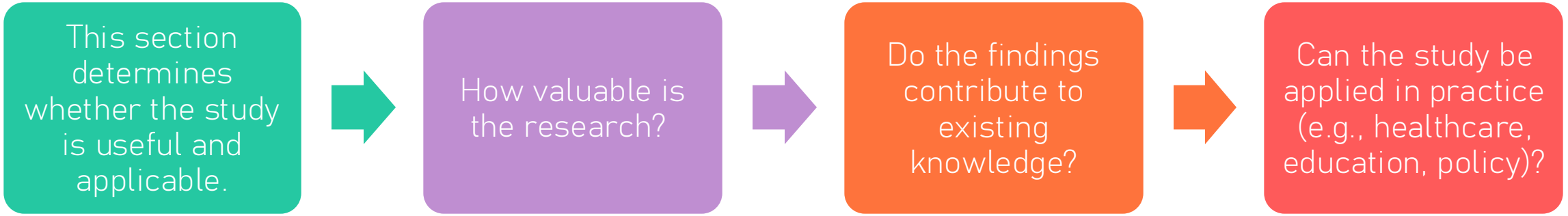
- This section assesses how well the findings are presented and analysed
- Have ethical issues been taken into consideration?
- Was informed consent obtained?
- Were ethical guidelines followed (e.g., confidentiality, data protection)?
- Was the data analysis sufficiently rigorous?
- Was the data analysed systematically?
- Were themes clearly identified?
- Is there a clear statement of findings?
- Were results clearly presented?
- Are conclusions backed by data?

Section B: What are the results?

Section B: What are the results?	
7. Have ethical issues been taken into consideration?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If there are sufficient details of how the research was explained to participants for the reader to assess whether ethical standards were maintained* • *If the researcher has discussed issues raised by the study (e.g. issues around informed consent or confidentiality or how they have handled the effects of the study on the participants during and after the study)* • *If approval has been sought from the ethics committee*	

8. Was the data analysis sufficiently rigorous?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If there is an in-depth description of the analysis process* • *If thematic analysis is used. If so, is it clear how the categories/themes were derived from the data* • *Whether the researcher explains how the data presented were selected from the original sample to demonstrate the analysis process* • *If sufficient data are presented to support the findings* • *To what extent contradictory data are taken into account* • *Whether the researcher critically examined their own role, potential bias and influence during analysis and selection of data for presentation*	
9. Is there a clear statement of findings?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If the findings are explicit* • *If there is adequate discussion of the evidence both for and against the researcher's arguments* • *If the researcher has discussed the credibility of their findings (e.g. triangulation, respondent validation, more than one analyst)* • *If the findings are discussed in relation to the original research question*	

CASP Section C: Will the results help locally?



Section C: Will the results help locally?	
10. How valuable is the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If the researcher discusses the contribution the study makes to existing knowledge or understanding (e.g., do they consider the findings in relation to current practice or policy, or relevant research-based literature* • *If they identify new areas where research is necessary* • *If the researchers have discussed whether or how the findings can be transferred to other populations or considered other ways the research may be used*	

Section C

- Will the results help locally?

Marking Criteria

Identify the chosen article (citation) States the research aim and importance (research gap) State the purpose of your critical appraisal and mention which CASP you are using	
Study Design: State why qualitative methodology is appropriate for the study's aim Discuss the rationale for design choice (e.g., phenomenology, grounded theory, ethnography) Demonstrate awareness of how methodology supports interpretive or experiential inquiry	10
Sampling and Recruitment Evaluate clarity of recruitment methods Discuss appropriateness of participant selection Identify any recruitment biases, gaps or transparency issues Note whether sample was justified (e.g., purposive, convenience, theoretical sampling)	10
Data Collection Methods Evaluates whether data collection effectively addresses the research question Considers justification of methods (e.g., interviews, focus groups, observations) Notes transparency of procedures (e.g., topic guides, field notes)	10
Data Analysis and Findings Evaluate whether researcher reflexivity is addressed Consider potential influence, bias, positionality or role negotiation Discuss researcher's impact on data collection and interpretation	10
Strengths, Limitations and Trustworthiness	10

Qualitative Articles Should Report:

- Relevance of question and rationale for approach
- Sampling strategy (selection, saturation)
- Data collection (interview guide, data)
- Analysis

Marking Criteria

Identifies gaps or unreported ethical processes	
Relevance to Physiotherapy Practice Discuss practical implications for physiotherapy Consider transferability and real-world application Reflect on how evidence supports evidence-based practiced	25
Conclusion Summarise the overall quality of the study Provide a final judgment on usefulness for practice	10
Reference and Quality of Writing Include references from peer-reviewed sources wherever possible. References and citations accurately follow LondonMet Harvard style An appropriate number of references are used to back up statements made in text. Uses a scientific, academic style of writing Template should be used for the coursework	10
	Total: /100

Marking Criteria

PT5053 – Combined Marking Framework (Quantitative & Qualitative)

Section	Weight (%)	80%+ (Excellent)	70–79% (Excellent)	60–69% (Good)	50–59% (Sound)	40–49% (Adequate)	30–39% (Inadequate)	<29% (Poor)
Introduction & Background	5%	Demonstrates an exceptional depth and thorough grasp of the clinical context; precise aim/question; article fully identified; explicit use of relevant CASP; sets clear scope.	Demonstrates a thorough and detailed understanding f the topic; clear objectives; minor gaps; article and CASP identified.	Demonstrates an excellent understanding of the topic; clear objectives but limited theoretical integration; article and CASP identified.	Demonstrates a sound understanding of the topic; objectives present but unfocused; article and CASP identified.	Demonstrates an adequate understanding of the topic, but imprecise objectives; weak coherence; article and CASP identified.	Demonstrates an insufficient understanding of the topic; unclear objectives; little grasp; may omit aim/CASP.	Demonstrates a very poor understanding of the topic; no clear objectives or relevant content; no aim/question or CASP.

Questions?



Practice

- Writing Questions: Qualitative Research Questions
- What is your topic of interest?
- Use the framework (PEO, Spider)
- Ensure a VERB is used when writing question

Received: 15 May 2024 | Revised: 3 June 2024 | Accepted: 6 June 2024

DOI: 10.1002/msc.1908

RESEARCH ARTICLE

WILEY

The experiences of patients with musculoskeletal conditions accessing first contact physiotherapy practitioner appointments in general practice in the UK: A qualitative study

Kirsten Lamb[1] | Christine Comer[1] | Nicola Walsh[2] | Julia Smith[1] | Krystal Tang[1] | Gretl McHugh[3]

Practice Article

Section A Are the results valid?	
1. Was there a clear statement of the aims of the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *what was the goal of the research?* • *why was it thought important?* • *its relevance*	
2. Is a qualitative methodology appropriate?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If the research seeks to interpret or illuminate the actions and/or subjective experiences of research participants* • *Is qualitative research the right methodology for addressing the research goal?*	
3. Was the research design appropriate to address the aims of the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *if the researcher has justified the research design (e.g., have they discussed how they decided which method to use)*	
4. Was the recruitment strategy appropriate to the aims of the research?	☐ Yes ☐ No ☐ Can't Tell
CONSIDER: • *If the researcher has explained how the participants were selected* • *If they explained why the participants they selected were the most appropriate to provide access to the type of knowledge sought by the study* • *If there are any discussions around recruitment (e.g. why some people chose not to take part)*	

CASP Section A Questions 1-6

5. Was the data collected in a way that addressed the research issue?	☐ Yes ☐ No ☐ Can't Tell

CONSIDER:

- *If the setting for the data collection was justified*
- *If it is clear how data were collected (e.g. focus group, semi-structured interview etc.)*
- *If the researcher has justified the methods chosen*
- *If the researcher has made the methods explicit (e.g. for interview method, is there an indication of how interviews are conducted, or did they use a topic guide)*
- *If methods were modified during the study. If so, has the researcher explained how and why*
- *If the form of data is clear (e.g. tape recordings, video material, notes etc.)*
- *If the researcher has discussed saturation of data*

6. Has the relationship between researcher and participants been adequately considered?	☐ Yes ☐ No ☐ Can't Tell

CONSIDER:

- *If the researcher critically examined their own role, potential bias and influence during (a) formulation of the research questions (b) data collection, including sample recruitment and choice of location*
- *How the researcher responded to events during the study and whether they considered the implications of any changes in the research design*

CASP Section A Questions 1-6